2.1 The encoders, and how they were compared

Every encoder in this document was scored on ONE basis, so the sheet's bars are comparable: a compound-cold split of the Vanacloig profiles (the compound is held out with every one of its doses; 32 compounds on each of three fold seeds, 96 compound-evaluations), the centered Spearman across strains between the predicted and the measured profile of the held-out compound, and the median over the evaluations as the score. Three kinds of encoder sit on that basis.

- **Fixed chemistry encoders with a shallow learner** (the ladder, 038): 15 ways of embedding a compound, four fingerprint families (FCFP4 and ECFP4 as counts or bits, MACCS keys), RDKit's 2D descriptors, seven pretrained molecular models (mol2vec, MolE, Uni-Mol, ChemBERTa-2 in both pretraining modes, MoLFormer-XL, a ZINC RoBERTa) and three concatenations, each run with kernel ridge and nearest neighbors under linear, RBF and Tanimoto kernels, 62 learner-by-encoder arms, every regularization chosen by a nested fold. The best is nested ridge on FCFP4 counts at 0.359; the best pretrained model is Uni-Mol v1 (3D) at 0.323; the lowest single encoder is ChemBERTa-2 MLM at 0.219.
- **Learned environment encoders** (038 rounds 9 and 10): the bilinear control (a compound embedding read against a learned gene embedding, no attention) and the environment encoder (the compound token with its dose in front of the gene tokens of the cell graph transformer), trained on the same folds, three seeds each; the encoder is read against its control paired on compound and fold, median difference -0.036, up on 37 of 96.
- **The mixture trainer** (040), the same transformer with the public screens and the host records in one objective, which scores on its own basis (the 28 compounds with a molar dose, folds 0 to 2): 0.309 for the gene readout and, under the cell readout, inside its fold noise (section 4).

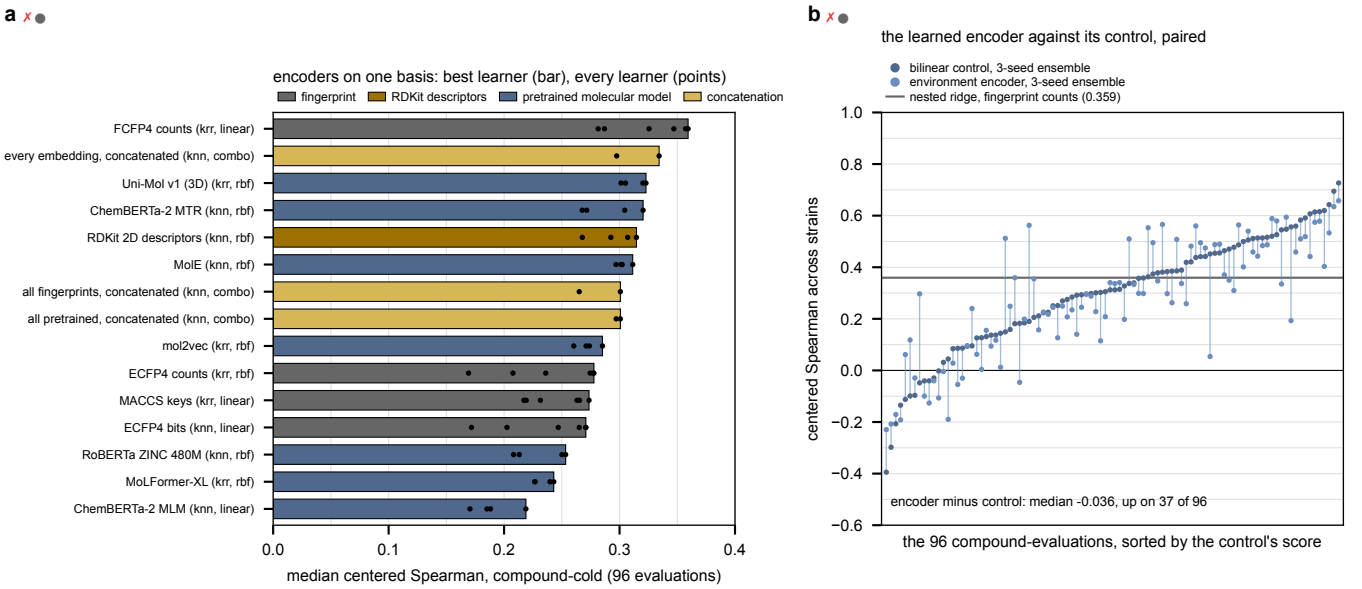


Figure 9: Supplement seven: the encoders compared. a, Every chemistry encoder of the ladder on the one basis, its best learner as the bar (the learner named in the label) and every learner it was run with as a point; color is the encoder family: fingerprints, descriptors, pretrained models, concatenations. Nested ridge on fingerprint counts leads; no pretrained model reaches it. **b**, The learned environment encoder against its bilinear control, paired on compound and fold over the 96 evaluations, sorted by the control's score; the line is nested ridge. The encoder is not above its control (median difference -0.036) and neither is above ridge. ✗●